

Piezoelectric Sensing of Mask Surface Waves for Noise-Suppressive Speech Input

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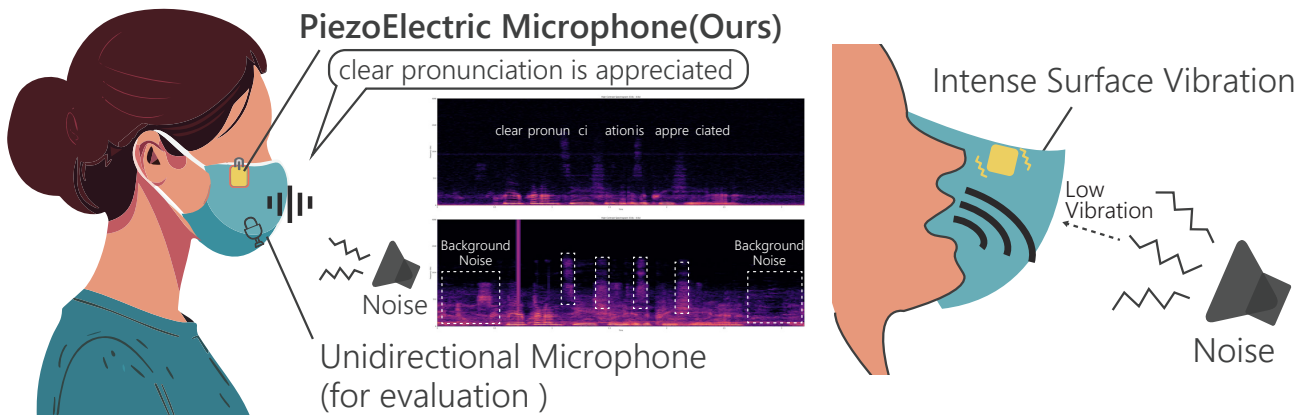


Figure 1: An overview of the detachable piezoelectric microphone, which can dominantly capture the mask surface vibration.

ABSTRACT

Voice-controlled medical devices in operating rooms face challenges due to face masks and noise, hindering speech recognition accuracy. We present a novel solution: a detachable piezoelectric microphone attached to a face mask, prioritizing hygiene and comfort. By capturing surface vibrations from the user's voice, our system significantly improves scale-invariant signal-to-distortion ratio (Si-SNR) compared to traditional unidirectional microphones, especially for whispered speech, as demonstrated in a simulated noisy environment. This real-time, noise-suppressive approach offers a promising avenue for enhancing voice input and conversations in various applications and settings.

CCS CONCEPTS

• **Human-centered computing** → *Mobile devices*; **Sound-based input / output**.

KEYWORDS

noise-suppressive microphone, speech enhancement, piezoelectric, wearable device, voice user interface

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1 INTRODUCTION

In modern healthcare environments, particularly operating rooms, there is a growing expectation for voice-controlled medical devices and automated record-keeping systems [14, 17, 22, 23].

However, the widespread adoption of face masks in these settings presents a significant challenge: masks muffle speech, degrading the quality of voice input [1, 16]. This issue is further compounded by the inherently noisy environment of medical facilities.

Noise suppression software [18] and multimodal speech enhancement [6] have been proposed. However, they often require substantial computational resources, making them impractical for resource-constrained devices. Earphone-based source separation methods [3, 19] can inadvertently treat frontal sound as noise. Mask-type microphones [9] pose hygiene concerns for long-term use due to their direct skin contact and difficulty in removal. Wearable silent speech solutions [5, 10, 12, 13] are difficult to apply to clear speech communication.

To overcome these limitations, we propose a novel approach: a detachable piezoelectric sensor that captures the user's voice through mask surface vibrations. This method builds upon the concept of throat microphones, which have been used in noisy environments [7, 20]. However, traditional throat microphones suffer from high-frequency attenuation and are susceptible to spike noise during movement [8].

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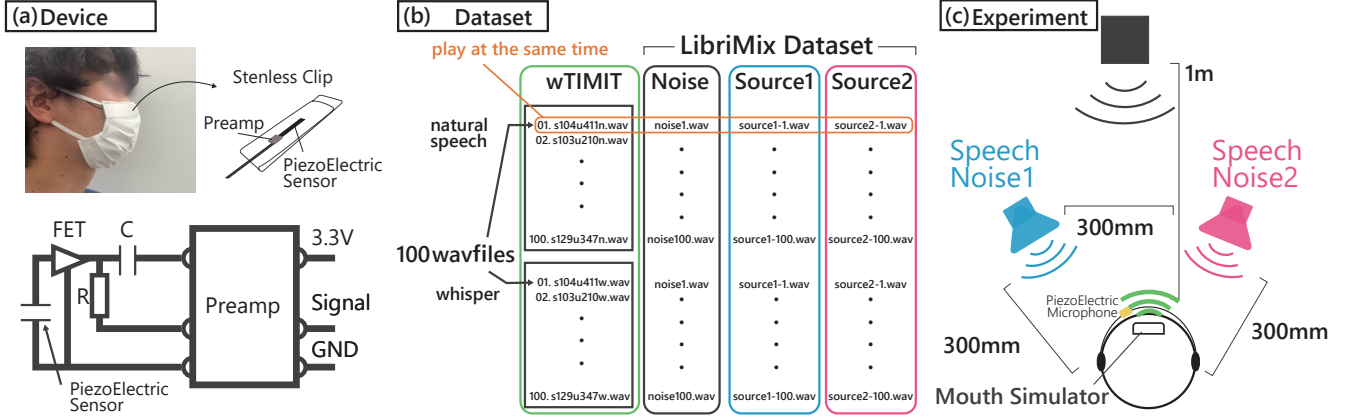


Figure 2: (a) Device: we implemented a detachable noise-suppressive microphone with a piezoelectric sensor. (b)(c) we use natural and whispered speech data from wTIMIT dataset [15] for simulating the user’s speech. Also, we used speech and noise clips from LibriMix [4] and Wham[21] for simulating noisy environment.

2 PROTOTYPING AND EVALUATION

We developed a prototype of a detachable piezoelectric microphone. The piezoelectric element utilized a biodegradable piezoelectric sensor[2], with voltage amplified with FETs and a preamp. The piezoelectric element was affixed to a stainless steel clip, which was then attached to a non-woven fabric mask (Figure 2(a)). This configuration allowed for vibration recording through the planar clip rather than the mask body itself.

For voice collection, we employed a Voice Simulator conforming to human mouth standards [11] to artificially reproduce speech. Recordings were made simultaneously using our proposed microphone (PiezoMic) and a unidirectional microphone (UniMic). To eliminate latency differences due to Bluetooth transmission, both microphones were directly connected to a computer (Apple Mac M3) via wired connections. We recorded two voice types (natural, and whispered speech) using 100 voice samples (Figure 2(a)) from the WTIMIT dataset [15], where the same speaker uttered identical content in both voice types.

To simulate noisy environments, we generated two types of noise from speech separation dataset [4, 21]: Speech Noise and Ambient Noise. These were positioned as shown in Figure 2(c), with each speaker’s output calibrated to 60 dB at the microphone position. All recordings were conducted in a quiet, non-reverberant environment with an ambient noise level of 30.0 dB.

3 RESULTS AND CONCLUSION

Table 1: The Speech Quality of Microphones (Mean $\pm$ SD)

	STOI $\uparrow$	PESQ $\uparrow$	SISNR $\uparrow$
UniMic Natural	0.238 $\pm$ 0.061	1.346 $\pm$ 0.285	-41.599 $\pm$ 11.996
UniMic Whisper	0.108 $\pm$ 0.037	1.311 $\pm$ 0.331	-50.628 $\pm$ 9.510
PiezoMic Natural	0.096 $\pm$ 0.047	1.834 $\pm$ 0.525	-27.531 $\pm$ 8.811
PiezoMic Whisper	0.037 $\pm$ 0.025	1.597 $\pm$ 0.384	-10.694 $\pm$ 8.653

We conducted a comparative analysis of our piezoelectric microphone (PiezoMic) against a unidirectional microphone (UniMic) in both quiet and noisy environments, as described above. The evaluation was based on three key metrics: Short-Time Objective Intelligibility (STOI), Perceptual Evaluation of Speech Quality (PESQ), and Scale-Invariant Signal-to-Noise Ratio (SI-SNR). Table 1 presents the results of this comparison.

Table 2: Pairwise t-test Results for Microphone Comparison

Metric	Speech Type	T-statistic	P-value	Effect Size
STOI	Natural	-18.600	2.497e-45	-2.631
	Whisper	-15.863	4.116e-37	-2.243
PESQ	Natural	8.177	3.438e-14	1.156
	Whisper	5.637	5.911e-08	0.797
SISNR	Natural	9.452	9.809e-18	1.337
	Whisper	31.060	4.805e-78	4.392

To assess the statistical significance of the differences between UniMic and PiezoMic across each metric, we conducted paired t-tests. The results of these tests are summarized in Table 2.

While the UniMic demonstrated superior performance in terms of Short-Time Objective Intelligibility (STOI), the PiezoMic showed advantages in both Perceptual Evaluation of Speech Quality (PESQ) and Scale-Invariant Signal-to-Noise Ratio (SI-SNR) measures.

Notably, the PiezoMic significantly outperformed the UniMic in SI-SNR, a metric that specifically represents the ratio of signal to noise. This superiority was observed for both natural and whispered speech conditions. The consistent advantage in SI-SNR suggests that the PiezoMic is particularly effective in noisy environments.

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